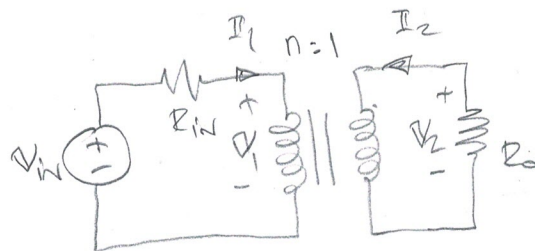


7.47)



$$R_{in} = 100 \Omega$$

$$R_o = 100 \Omega$$

$$V_{in} = V_{pk} \cos(\omega t) \text{ volt}$$

Så att $n=1$ så att man får maximal effektoverföring.

Lösning: VIETATT

$$Z_1 = \frac{1}{n^2} Z_2 = M^2 Z_2$$

$$\Rightarrow Z_1 = M^2 R_o$$

STRÖMMENGES AV KSP

$$-V_{in} + (R_{in} + Z_1) I_1 = 0$$

$$I_1 = \frac{V_{in}}{R_{in} + Z_1} = \frac{V_{in}}{R_{in} + M^2 R_o} \quad (1)$$

STRÖM PÅ SEKUNDÄRSIDAN

$$I_2 = M I_1 \quad (2)$$

SPÄNNING PÅ SEKUNDÄRSIDAN

$$V_2 = \frac{1}{M} V_1 \quad (3)$$

ÖVERFÖRD EFFEKT

$$S_2 = \frac{1}{2} V_2 I_2^* = \frac{1}{2} \frac{1}{M} V_1 M I_1^* =$$

$$= \frac{1}{2} V_1 \frac{V_{in}^*}{R_{in} + M^2 R_o}$$

$$\text{MEN } V_1 = V_{in} - R_{in} I_1$$

$$S_2 = \frac{1}{2} \frac{V_{in} - R_{in} I_1}{R_{in} + M^2 R_o} V_{in}^* =$$

$$= \frac{1}{2} \left[\frac{V_{pk}^2}{R_{in} + M^2 R_o} - \frac{R_{in} V_{pk}^2}{(R_{in} + M^2 R_o)^2} \right] =$$

$$= \frac{1}{2} \frac{V_{pk}^2}{R_{in} + M^2 R_o} \left[1 - \frac{R_{in}}{R_{in} + M^2 R_o} \right]$$

☺

$$\begin{aligned} \beta_2 &= \frac{1}{2} \frac{V_{PK}^2}{R_{IN} + M^2 R_0} \left[\frac{R_{IN} + M^2 R_0 - R_{IN}}{R_{IN} + M^2 R_0} \right] = \frac{1}{2} \frac{V_{PK}^2}{R_{IN} + M^2 R_0} \frac{M^2 R_0}{R_{IN} + M^2 R_0} \\ &= \frac{1}{2} \frac{V_{PK}^2 M^2 R_0}{(R_{IN} + M^2 R_0)^2} = \frac{1}{2} V_{PK}^2 R_0 f(M) \end{aligned}$$

$$\text{Nur } f(M) = \frac{M^2}{(R_{IN} + M^2 R_0)^2}$$

Derivens für ein Maximum

$$\frac{df}{dM} = \frac{2M(R_{IN} + M^2 R_0)^2 - M^2 \cdot 2(R_{IN} + M^2 R_0) \cdot 2MR_0}{(R_{IN} + M^2 R_0)^4} = 0$$

$$\Rightarrow 2M(R_{IN} + M^2 R_0)^2 - M^2 \cdot 2(R_{IN} + M^2 R_0) \cdot 2MR_0 = 0$$

$$R_{IN} + M^2 R_0 = 2M^2 R_0$$

$$R_{IN} = M^2 R_0$$

$$M^2 = \frac{R_{IN}}{R_0}$$

$$M = \sqrt{\frac{R_{IN}}{R_0}} = \sqrt{\frac{100}{1200}} = \underline{\underline{0.2887}}$$

VAD für VIDA?

$$\beta_2 = \frac{1}{2} \frac{V_{PK}^2 \frac{R_{IN}}{R_0} R_0}{(R_{IN} + \frac{R_{IN}}{R_0} R_0)^2} = \frac{1}{2} \frac{V_{PK}^2 R_{IN}}{4R_{IN}} = \underline{\underline{\frac{1}{8} V_{PK}^2}}$$